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Method for making a three-dimensional liquid crystal polymer electronic device

Abstract

A method is for making an electronic device that includes a multilayer circuit board having a non-planar three-dimensional shape defining a membrane switch recess therein. The multilayer circuit board may include at least one liquid crystal polymer (LCP) layer, and at least one electrically conductive pattern layer thereon defining at least one membrane switch electrode adjacent the membrane switch recess to define a membrane switch. The electronic device may further include a compressible dielectric material filling the membrane switch recess. The electronic device may also include at least one spring member within the membrane switch recess.

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to the field of circuit boards, and, more particularly, to circuit boards including a membrane switch provided therein and related methods.

BACKGROUND OF THE INVENTION

(2) An electronic device may include one or more circuit boards. A typical circuit board is a two-dimensional (2D) planar board that mechanically supports electronic components. The electronic components may comprise, for example, resistors, capacitors, switches, batteries, and other more complex integrated circuit components, i.e. microprocessors. The circuit board typically comprises a dielectric material, for example, a plastic material.

(3) The circuit board may include conductive traces on the surface for connecting the electronic components to each other. As electronic circuitry has become more complex, multilayer circuit boards with at least two electrically conductive pattern layers have been developed. Typically, the different conductive trace layers of a multilayer circuit board may be connected through vertically extending vias, which comprise conductive materials, for example, metal. A typical multilayer circuit board may comprise a plurality of core layers with bonding layers therebetween affixing the adjacent core layers together. Each core layer typically includes a dielectric layer with electrically conductive pattern layers on the opposing surfaces of the dielectric layer. Typically, during manufacture of the multilayer circuit boards, the core and bonding layers are stacked together and then heated (laminated) to cause the bonding layer to affix the adjacent core layers together.

(4) Even with the advent of the multilayer circuit board, as the mounted circuitry has become even more complex, the size of the circuit board and associated packaging has also increased. This increase in size may pose installation drawbacks in applications where space may be limited or where fitting a planar two-dimensional circuit board may be problematic. Three-dimensional (3D) circuit boards are an approach to this drawback of typical 2D planar circuit boards. As with the typical planar multilayer circuit board, the typical 3D circuit board may comprise a plurality of core layers with bonding layers therebetween affixing adjacent layers together.

(5) Advantageously, 3D circuit boards may perform functions beyond the traditional mechanical support and electrical connection functions of the 2D circuit board. In other words, the 3D circuit board may be a multifunctional structure. For example, the 3D circuit board may perform mechanical, aerodynamic, and encapsulation functions.

(6) Another approach to growth in circuit board size is integrating external electronic components into the circuit board, for example, batteries, and switches. For example, U.S. Pat. No. 7,045,246 to Simburger et al. discloses a thin film battery embedded in a multilayer thin film flexible circuit board. The circuit board comprises polyimide material, which may have some undesirable material characteristics.

(7) One method to forming 3D circuit boards is disclosed in U.S. Pat. No. 8,161,633 issued Apr. 24, 2012, to Shacklette et al., also assigned to the assignee of the present invention, which is incorporated in its entirety by reference. The method includes thermoforming core layers individually on a 3D mold structure, stacking the thermoformed core layers, and laminating the stacked thermoformed layers at even a greater temperature. One possible drawback of this method is the two-step heating and cooling process increases manufacturing time and limits productivity.

SUMMARY OF THE INVENTION

(8) In view of the foregoing background, it is therefore an object of the present invention to provide an electronic device with a multilayer circuit board including a membrane switch therein having effective sealing and good electrical properties.

(9) This and other objects, features, and advantages in accordance with the present invention are provided by an electronic device including a multilayer circuit board having a non-planar three-dimensional shape defining a membrane switch recess therein. The multilayer circuit board may include at least one liquid crystal polymer (LCP) layer, and at least one electrically conductive pattern layer thereon defining at least one membrane switch electrode adjacent the membrane switch recess to define a membrane switch. The electronic device may further include a compressible dielectric material filling the membrane switch recess. The electronic device may also include at least one spring member within the membrane switch recess. Advantageously, the multilayer circuit board may provide a hermetic seal for the membrane switch recess

therein.

(10) Additionally, the electronic device may further comprise circuitry carried by the multilayer circuit board and being coupled to the membrane switch. The at least one LCP layer may further comprise at least one pair thereof, and the multilayer circuit board may further comprise a bonding layer between the at least one pair of LCP layers. In certain embodiments, the bonding layer may comprise a curable bonding layer. In other embodiments, the bonding layer may comprise a thermoplastic bonding layer.

(11) Another aspect is directed to a method for making an electronic device comprising forming a multilayer circuit board having a non-planar three-dimensional shape defining a membrane switch recess therein. The multilayer circuit board may comprise at least one liquid crystal polymer (LCP) layer, and at least one electrically conductive pattern layer thereon defining at least one membrane switch electrode adjacent the membrane switch recess to define a membrane switch of the electronic device. The electrically conductive pattern layer may comprise at least one of copper, nickel, silver, gold, indium, lead, tin, carbon, and aluminum.

(12) Moreover, the method may further comprise mounting circuitry on the multilayer circuit board, the circuitry being coupled to the membrane switch. The at least one LCP layer may comprise at least one pair thereof, and the multilayer circuit board may further comprise a bonding layer between the at least one pair of LCP layers.

(13) Furthermore, the forming of the multilayer circuit board may comprise forming a stacked arrangement comprising at least one pair of LCP layers with a bonding layer therebetween. The forming of the multilayer circuit board may also comprise heating and applying pressure to the stacked arrangement to shape the stacked arrangement into a non-planar three-dimensional shape and concurrently causing the bonding layer to bond together the adjacent LCP layers of the stacked arrangement. Each of the LCP layers may have a melting temperature, and the bonding layer may have a bonding temperature less than the melting temperature of each of the LCP layers.

(14) In certain embodiments, the bonding layer may comprise a curable bonding layer, and the bonding temperature may comprise a curing temperature for the curable bonding layer. In other embodiments, the bonding layer may comprise a thermoplastic bonding layer, and the bonding temperature may comprise a melting temperature for the thermoplastic bonding layer. The forming of the stacked arrangement may comprise initially forming a stacked planar arrangement.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a flowchart illustrating a method for making a non-planar 3D multilayered circuit board according to the present invention.

(2) FIG. 2 is a perspective view of a non-planar 3D multilayer circuit board made according to the method of FIG. 1.

(3) FIG. 3 is a schematic cross-sectional diagram of an electronic device according to the present invention.

(4) FIG. 4 is a flowchart illustrating a method for making the electronic device of FIG. 3.

(5) FIGS. 5-12 illustrate an embodiment of the method for making the electronic device of FIG. 3.

(6) FIG. 13 is an isometric view from the top of another electronic device according to the present invention.

(7) FIG. 14 is a second isometric view from the bottom of the electronic device of FIG. 13.

(8) FIG. 15 is a cross-section view of the electronic device of FIG. 13.

(9) FIG. 16 is a flowchart illustrating a method for making the electronic device of FIG. 13.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(10) The present invention will now be described more fully hereinafter with reference to the accompanying drawings, in which preferred embodiments of the invention are shown. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art. Like numbers refer to like elements throughout.

(11) Referring initially to FIGS. 1-2, a flowchart 10 illustrates a method for making a non-planar three-dimensional (3D) multilayer circuit board 80. From the start (Block 11), the method illustratively includes forming (Block 12) a electrically conductive pattern layer 83 on inner surfaces of liquid crystal polymer

(LCP) layers **82, 85**, and forming (Block **13**) a stacked arrangement, which may be initially planar, the stacked arrangement comprising at least one pair of the LCP layers **82, 85** with a bonding layer **84** therebetween.

(12) The electrically conductive pattern layer **83** is illustratively formed on each LCP layer **82, 85**. As will be appreciated by those skilled in the art, the electrically conductive pattern layer **83** may be stripped thereafter. In some embodiments, the electrically conductive pattern layer **83** may be formed on a single LCP layer **82, 85**. As will be appreciated by those skilled in the art, the multilayer circuit board **80** may be defined by the number of the electrically conductive pattern layers **83** thereon.

(13) The LCP layers **82, 85** may comprise, for example, Rogers F/Flex® 3600, 3850 core material layers or Nippon Steel Espanex L, Std-Type core material layers. Each of the LCP layers may comprise a biaxially oriented LCP layer. Advantageously, the biaxially oriented LCP layers have low values for the X and Y coefficients of thermal expansion (CTE) and relatively high values for the Z CTE. For example, Rogers F/Flex® 3600 and 3850 both have X, Y, and Z values for CTE of 17 (10.sup.-6*1/° C.), 17 (10.sup.-6*1/° C.), and 150 (10.sup.-6*1/° C.), respectively.

(14) Advantageously, LCP has electrical properties that may helpful for use in the 3D multilayer circuit **80** board. For example, Rogers F/Flex® 3600 and 3850 both have a low dielectric constant of 2.9 and a loss tangent at 10 GHz of 0.0025. Moreover, LCP (Rogers F/Flex® 3600/3850 and or Nippon Steel Espanex L) has hermetic properties and a low water uptake of 0.04%, and a Young modulus of in the range of 2400-3000 MegaPascals. Advantageously, LCP provides a mechanically robust dielectric material. Moreover, the low loss tangent provides for lower losses in high frequency circuitry, and the lower dielectric constant provides the ability to reduce line spacing and create more compact circuit layouts.

(15) The stacked arrangement illustratively includes the electrically conductive pattern layer **83**, for example, metal traces, on each of the LCP layers **82, 85**. The electrically conductive pattern layer **83** may comprise at least one of copper, nickel, silver, gold, indium, lead, tin, carbon, and aluminum or an alloy thereof. For example, the electrically conductive pattern layer **83** may comprise a base metal layer of one type and a second metal layer of a second type thereon, in other words, a multilayer composite.

(16) The electrically conductive pattern layer **83** may be applied to some or all of the LCP layers **82, 85** before or after the thermoforming and lamination step. As will be appreciated by those skilled in the art, applying the electrically conductive pattern layer **83** to the inner surfaces of the LCP layers **82, 85** may need to be performed before the thermoforming and lamination step.

(17) In certain 3D forms with high aspect ratios, forming the electrically conductive pattern layers **83** before the thermoforming and lamination step may be more difficult. In these high aspect ratio 3D circuit boards, the electrically conductive pattern layers **83** may be formed thereafter using, for example, inkjet printing or silk screening. As will be appreciated by those skilled in the art, the electrically conductive pattern layer **83** may comprise any material with suitable conductivity properties. Advantageously, the low value of X-Y CTE and high value of X-Y tensile modulus for the LCP layers may prevent breaks and discontinuities in the electrically conductive pattern layer **83** during the thermoforming and lamination step. Indeed, the linear (X-Y) CTE of copper is 17 (10.sup.-6*1/° C.), which advantageously matches the X-Y CTE of the LCP layers.

(18) Further advantage stems from the thermoforming step being carried out at a temperature that is significantly below the melting point and approximately equal to or even as much as 30° C. below the glass transition temperature of the LCP core layers. Under such conditions, the LCP core layers retain a modulus significantly higher than that of the bonding layer **84**, a condition that will act to limit and more uniformly spread the deformation of the LCP layers **82, 85** and the copper traces thereupon, thereby reducing the chances for a break in the copper traces **83** caused by excessive elongation.

(19) The method illustratively includes heating and applying pressure to the stacked arrangement to shape the stacked arrangement into a non-planar 3D shape and concurrently causing (Block **15**) the bonding layer **84** to bond together the adjacent LCP layers **82, 85** of the stacked arrangement to thereby form the non-planar 3D multilayer circuit board. The method ends at (Block **16**), an exemplary multilayer circuit board **80** with circuitry **81** thereon being shown in FIG. **2**. In other words, the multilayer circuit board is thermoformed and laminated in one step.

(20) Additionally, the heating (Block **15**) may comprise heating in a range of 170 to 230° C. Preferably, the heating (Block **15**) may comprise heating in a range of 180 to 220° C. As will be appreciated by those skilled in the art, the low temperature bound is determined by the respective temperature that provides adequate plasticity in the LCP layers **82, 85** for thermoforming. In other words, the LCP should be flexible enough to be shaped or have sufficient plasticity to deform under pressure or vacuum. The high temperature bound is

determined by the respective temperature that generates excessive plasticity or fluidity in the LCP layers **82**, **85**, thereby causing the LCP layers to excessively and/or unevenly deform during thermoforming. The high temperature limit may depend on the particular choice of LCP material, since the glass transition temperature and the melting point generally vary with the particular grade or manufacturer. It is generally preferred to perform the process at the lowest temperature that produces the desired permanent shape. A preferred temperature range is from about 180° C. to 220° C. The heating (Block **15**) may further comprise increasing the temperature at a constant rate and subsequently decreasing the temperature at a constant rate, for example, 5° C. per minute and 10° C. per minute, respectively.

(21) Applying pressure (Block **15**) may comprise applying pressure using at least one of a vacuum bag and a press mold, as will be appreciated by those skilled in the art. The pressure range may be 50-200 pounds per square inch (psi), for example. Preferably, the pressure range is about 80-120 psi. As will be appreciated by those skilled in the art, performing the concurrent thermoforming and lamination step at the low pressure bound may require a greater process temperature, and performing the concurrent thermoforming and lamination step at the higher pressure bound may require a lower process temperature.

(22) Moreover, each of the LCP layers **82**, **85** has a glass transition temperature and a melting temperature above the glass transition temperature. Near or above the glass transition temperature, the LCP layers **82**, **85** have a plasticity value for permitting thermoforming. The bonding layer **84** has a bonding temperature that is significantly below the melting point and approximately equal to or even slightly below the glass transition temperature of the LCP layers **82**, **85**. Advantageously, the steps of bonding/laminating the bonding layer and thermoforming the 3D multilayer circuit board **80** may be concurrently performed since the process temperature of the bonding layer **84** and the glass transition temperature of the LCP layers are within range.

(23) As will be appreciated by those skilled in the art, the layers of the 3D multilayer circuit board **80** may be precisely aligned to fit circuit board features of one layer to the appropriate features in adjacent layers. For example, the layers may have alignment holes drilled in them before/after the thermoforming and lamination step, the holes being aligned with posts in the press mold to be used in either a mechanical press or within a vacuum bag subject to heat and pressure within an autoclave.

(24) In some embodiments, the bonding layer **84** may comprise a curable bonding layer, and the bonding temperature may comprise a curing temperature for the curable bonding layer. For example, the bonding layer **84** may comprise a Bismaleimide-Triazine (BT) resin. For example, in these embodiments, the bonding layer may comprise Gore Speedboard® C/LF preimpregnated thermoset bonding layers. The process/curing temperature of the Gore Speedboard® C/LF preimpregnated thermoset bonding layers is recommended to be about 180-220° C. As will be appreciated by those skilled in the art, other curable bonding layers having a process/curing temperature within range of the glass transition temperature of LCP may be used.

(25) In other embodiments, the bonding layer **84** may comprise a thermoplastic bonding layer, and the bonding temperature may comprise a melting temperature for the thermoplastic bonding layer. Furthermore, the bonding layer **84** may comprise, for example, a thermoplastic, such as, chlorotrifluoroethylene (CTFE). For example, in these embodiments, the thermoplastic bonding layer may comprise Arlon® 6250 bonding layers. The process/melting temperature of the Arlon® 6250 bonding layers is within the range of 120-150° C. As will be appreciated by those skilled in the art, other thermoplastic bonding layers, such as, Arlon® 6700 or Rodgers® 3001, having a process/melting temperature within range of the glass transition temperature of LCP may be used.

(26) Referring now to FIG. **3**, an exemplary electronic device **20** is now described. The electronic device **20** illustratively includes a multilayer circuit board **27** having a non-planar three-dimensional shape defining a battery component receiving recess **32** of the electronic device therein. The battery component receiving recess **32** may receive, for example, active materials, electrolytes **22**, spacers, an anode, a cathode, and current collectors. The LCP layers **23** may function to provide environmental packaging as well as a substrate for circuitry **21** that also comprises the electronic device **20**.

(27) As will be appreciated by those skilled in the art, the non-planar three-dimensional shape defining a battery component receiving recess **32** therein may be manufactured using the method for making a non-planar 3D multilayered circuit board described above. Alternatively, other methods of thermoforming may be used as will also be appreciated by those skilled in the art, for example, the two step lamination and thermoforming process disclosed in U.S. Pat. No. 8,161,633 issued Apr. 24, 2012, to Shacklette et al.

(28) The multilayer circuit board **27** illustratively includes a plurality of LCP layers **23**, and a plurality of electrically conductive pattern layers **26** on some of or all of the LCP layers. The electrically conductive pattern layers **26** define a pair of battery electrodes (contacts) **31**, **30** adjacent the battery component

receiving recess **32**. The electrically conductive pattern layers **26** may comprise at least one of copper and aluminum, for example. More specifically, the battery contacts **31**, **30** include a cathode contact **30** comprising aluminum and an anode contact **31** comprising copper. As will be appreciated by those skilled in the art, other conductive metals may be used.

(29) In some embodiments (not shown), the multilayer circuit board may include a single LCP layer, and a metal foil layer thereon sealing the battery component receiving recess and the components contained therein. The metal foil layer may comprise, for example, gold, copper, nickel, iron, cobalt, aluminum, molybdenum, silver, zinc, titanium, and alloys thereof. Preferably, the metal foil may comprise copper, aluminum or stainless steel, or one of those metals plated or coated by a second metal.

(30) The electronic device **20** may further include anode and cathode active materials, an insulating spacer, optional metal current collectors, and a battery electrolyte **22** within the battery component receiving recess **32**. The battery electrolyte **22** may contact the battery electrode contacts **31**, **30** to define a battery, and with the multilayer circuit board **27** defining exterior portions for the battery. Moreover, the battery electrolyte **22** may comprise lithium ion electrolyte, for example.

(31) Advantageously, the electrolyte receiving recess **32** may define the boundaries of the battery components, for example, the electrolyte **22**, the anode and cathode active materials, the insulating spacer, the metal current collectors. In other words, the bare battery electrolyte **22** and other battery components may be integrated into the multilayer circuit board **27** without the typical packaging, for example, foil packaging. In other embodiments, the battery electrolyte **22** and other battery components may be integrated into the multilayer circuit board **27** with the typical packaging. The components of the battery, such as, current collectors, electrodes, and spacers, may be stacked between the LCP layers **23** prior to lamination, and the subsequent lamination step may both form the LCP around the battery stack and both laminate the multilayer circuit board **27** and seal the battery components in one step.

(32) As will be appreciated by those skilled in the art, the battery electrolyte **22** may comprise other electrolyte types. Moreover, the electrolyte **22** may be inserted into the battery component receiving recess **32** after thermoforming and lamination of the LCP layers **23**, for example, using an opening **28** (FIGS. 6-12) in the battery component receiving recess. Alternatively, the electrolyte packaged cell may be used and inserted before finishing the battery component receiving recess **32**. The battery electrolyte **22** may comprise a solid electrolyte or alternatively liquid electrolyte. In embodiments of the electronic device **20** where the non-planar three-dimensional shape defining the battery component receiving recess **32** is manufactured using the method for making a non-planar 3D multilayered circuit board described above, a solid electrolyte may be used and positioned in the LCP layers **23** before lamination. As will be appreciated by those skilled in the art, the battery electrolyte **22** may be stable at the thermoforming temperature. Alternatively, if using a liquid electrolyte, the LCP layers **23** may need to be laminated prior to injection through the opening **28** of the liquid electrolyte due to the high temperature of the thermoforming process and the likely instability of the liquid electrolyte during the thermoforming process.

(33) Additionally, the electronic device **20** illustratively includes circuitry **21** carried by the multilayer circuit board **27** and receiving power from the battery, which may be a rechargeable battery or a one-time use battery. The circuitry **21** may comprise, for example, passive components, display components, or/and active components, such as, an integrated circuit, etc. The multilayer circuit board **27** illustratively includes a bonding layer **25** between the LCP layers **23**. In certain embodiments, the bonding layer **25** may comprise a curable bonding layer. In other embodiments, the bonding layer **25** may comprise a thermoplastic bonding layer. The electronic device also includes illustratively a perimeter seal **24**, which may be the same as or different from the bonding layer **25**, but may be processed within the same temperature window that allows the lamination and the shaping of the electronic device **20**.

(34) Referring now also to FIG. 4, a flowchart **33** illustrates a method for making an electronic device **20**. From the start (Block **34**), the method illustratively begins with forming electrically conductive pattern layers **26** on inner surfaces of the LCP layers **23** (Block **36**). The method also includes forming a stacked arrangement, which may be initially planar, the stacked arrangement comprising at least one pair of LCP layers **23** with a bonding layer **25** therebetween (Block **37**), and positioning battery components within the LCP layers in alignment with the inner conductive pattern layers **26** (Block **38**). The contacts of the electronic device **20** may be coated with solder or conductive adhesives, such that the, solders may melt or the adhesives may cure during the lamination and thermoforming step.

(35) The method illustratively includes heating and applying pressure to the stacked arrangement to shape the stacked arrangement into a non-planar 3D shape and concurrently causing (Block **41**) the bonding layer

25 to bond together the adjacent LCP layers 23 of the stacked arrangement to thereby form a battery component receiving recess 32, in other words, forming a multilayer circuit board 27 having a non-planar three-dimensional shape defining a battery component receiving recess therein. (Block 41) The multilayer circuit board 27 includes LCP layers 23, and a plurality of electrically conductive pattern layers 26 thereon defining a plurality of battery electrodes 30, 31 adjacent the battery component receiving recess 32.

(36) The method also includes positioning (Block 42) a battery electrolyte 22 within the battery component receiving recess 32 and contacting the battery electrodes 30, 31 to define a battery for the electronic device. As will be appreciated by those skilled in the art, the battery electrolyte 22 may be inserted into the battery component receiving recess 32 further upstream, for example, by thermoforming around a prepackaged solid electrolyte cell or by including a solid or gel electrolyte among the battery components that can withstand the temperature and pressure of the lamination cycle. The method also illustratively includes mounting (Block 43) circuitry 21 on the multilayer circuit board 27 to receive power from the battery, the method ending at (Block 44).

(37) In another embodiment of the method, a non-planar 3D shape would be created during the process of (Block 41) using appropriate tooling during this combined thermoforming and lamination step. The process step of (Block 41) would then be followed by inserting a pre-packaged battery or full set of battery components into the battery component receiving recess 32 (Block 42) and sealing the battery into the non-planar 3D shape by applying an additional LCP layer 23 held in place by a perimeter seal 24. As will also be appreciated by those skilled in the art, the method described for embedding a battery can be applied to embedding similarly shaped objects, such as, a metal heat spreader or an integrated circuit.

(38) As will be appreciated by those skilled in the art, an exemplary implementation of the method of making the electronic device 20 follows. Referring additionally to FIGS. 5-12, the method is illustrated. Referring specifically to FIG. 6, the vias are formed in the top LCP layer 23, one via forming the opening 28 for subsequent injection of a liquid electrolyte 22.

(39) In FIG. 7, a slurry coat is applied and the cathode 30 and the anode 31 are formed out of aluminum and copper cladding, respectively. A lithium cobalt oxide electrode 29 is also formed adjacent the cathode 30. A graphite electrode 90 is also formed respectively adjacent the anode 31. The electrodes are then laminated (FIG. 8) and the battery is stacked-up. Referring specifically to FIG. 9, the top LCP layer 23 is thermoformed to form the battery component receiving recess 32. Referring specifically to FIG. 10, the perimeter seal 24 is laminated on the cathode 30. In FIG. 11, the electrolyte 22, for example, a liquid free gel-polymer electrolyte layer, is filled post thermoforming. The opening is also filled 28 to provide a seal. In FIG. 12, the vias are either hand painted or plated to tie in battery power.

(40) Referring now to FIGS. 13-15, another exemplary electronic device 45 is illustrated and illustratively includes a multilayer circuit board 53 having a non-planar three-dimensional shape. The non-planar three-dimensional shape defines a membrane switch recess 52 therein. As will be appreciated by those skilled in the art, the non-planar three-dimensional shape defining the membrane switch recess 52 therein may be manufactured using the method for making a non-planar 3D multilayered circuit board described above. Alternatively, other methods of thermoforming may be used as will also be appreciated by those skilled in the art, for example, the two step lamination and thermoforming process disclosed in U.S. patent application Ser. No. 11/695,685 to Shacklette et al.

(41) The multilayer circuit board 53 illustratively includes a pair of LCP layers 50, 51, and a pair of conductive pattern layers 46, 47 thereon defining a plurality of membrane switch electrodes adjacent the membrane switch recess 52 to define a membrane switch. Advantageously, although typical membrane switch electrodes are plated in gold or nickel to prevent corrosion, since the thermoformed LCP layers 50, 51 provide hermetic or near hermetic properties, the membrane switch electrodes may comprise bare copper with no plating, thereby reducing manufacturing cost and durability.

(42) The electrically conductive pattern layers 46, 47 may comprise at least one of copper, nickel, silver, gold, indium, lead, tin, carbon, and aluminum or an alloy thereof. For example, the electrically conductive pattern layers 46, 47 may comprise a base metal layer of one type and a second metal layer of a second type thereon, in other words, a multilayer composite.

(43) Since it may be helpful to keep the membrane switch recess 52 open during the thermoforming and lamination step, it may be preferred that the lamination be made with a mechanical press that includes a top plate or tooling piece that is machined to produce the desired 3D form of the LCP layer 50. It may be further preferred that the top plate or tool exert a vacuum on the top surface of the LCP layer 50 to ensure that it conforms to the shape of the tooling during the forming and lamination step.

(44) The electronic device **45** may further include a compressible dielectric material filling the membrane switch recess **52**. For example, the dielectric filling material may comprise an air pocket. The electronic device **45** may also include at least one spring member **56** within the membrane switch recess **52**. Alternatively, certain embodiments may omit the spring member **56** since the thermoformed LCP layers **50**, **51** are mechanically elastic and resilient and may return to preformed 3D shape after the membrane switch recess **52** is depressed.

(45) Additionally, the electronic device **45** illustratively includes circuitry **54** carried by the multilayer circuit board **53** and being coupled to the membrane switch. Advantageously, the membrane switch may be coupled to and control the circuitry **54** integrated on the multilayer circuit board **53**. The multilayer circuit board **53** includes a bonding layer **55** between the LCP layers **50**, **51**.

(46) Referring now additionally to FIG. **16**, a flowchart **60** illustrates a method for making an electronic device **45**. From the start (Block **61**), the method illustratively begins with forming electrically conductive pattern layers **46**, **47** on inner surfaces of LCP layers **50**, **51** to define at least one pair of switch electrodes (Block **62**). The method also includes forming a stacked arrangement, which may be initially planar, the stacked arrangement comprising at least one pair of patterned LCP layers **50**, **51** with a bonding layer **55** therebetween (Block **63**). The stacked arrangement includes at least one electrically conductive pattern layer **46**, **47** on each of the LCP layers **50**, **51**.

(47) The method also includes heating and applying pressure and a selective vacuum to the stacked arrangement, and shaping (Block **65**) the stacked arrangement into a non-planar 3D shape and concurrently causing the bonding layer **55** to bond together the adjacent LCP layers **50**, **51** of the stacked arrangement to thereby form the 3D multilayer circuit board **53**. In other words, the method includes forming a multilayer circuit board **53** having a non-planar three-dimensional shape defining a membrane switch recess **52** therein.

(48) The method illustratively includes forming (Block **66**) electrically conductive pattern layers **46**, **47** on at least one of the outer surfaces of the LCP layers **50**, **51** for defining at least one membrane switch electrode adjacent the membrane switch recess **52** to define a membrane switch, i.e. to complete the circuit interconnects and optionally to connect to any surface mount components or circuitry **54**. As will be appreciated by those skilled in the art, the electrically conductive pattern layers **46**, **47** may be formed further upstream in the method, for example, during the process at Block **62**. Moreover, the method illustratively includes mounting the circuitry **54** (Block **70**) on the multilayer circuit board **53**, the circuitry being coupled to the membrane switch. The method ends at (Block **71**).

(49) Many modifications and other embodiments of the invention will come to the mind of one skilled in the art having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is understood that the invention is not to be limited to the specific embodiments disclosed, and that modifications and embodiments are intended to be included within the scope of the appended claims.

Claims

1. A method for making an electronic device comprising: forming a multilayer circuit board having a non-planar three-dimensional shape defining a membrane switch recess therein filled with a compressible dielectric material, the multilayer circuit board comprising at least one liquid crystal polymer (LCP) layer, and at least one electrically conductive pattern layer thereon defining at least one membrane switch electrode adjacent the membrane switch recess to define a membrane switch; wherein forming the multilayer circuit board comprises forming a stacked arrangement comprising at least one pair of LCP layers with a bonding layer therebetween, and heating the stacked arrangement in a range of 170° C. to 230° C. to shape the stacked arrangement into the non-planar three-dimensional shape and concurrently causing the bonding layer to bond together the adjacent LCP layers of the stacked arrangement.
2. The method according to claim 1 comprising positioning at least one biasing member in the membrane switch recess.
3. The method according to claim 2 wherein the at least one biasing member comprises at least one spring member.
4. The method according to claim 1 further comprising mounting circuitry on the multilayer circuit board, the circuitry being coupled to the membrane switch electrode.
5. The method according to claim 1 wherein each of the LCP layers has a melting temperature; and wherein the bonding layer has a bonding temperature less than the melting temperature of each of the LCP layers.

6. The method according to claim 1 wherein the bonding layer comprises a curable bonding layer; and wherein the bonding temperature comprises a curing temperature for the curable bonding layer.
 7. The method according to claim 1 wherein the bonding layer comprises a thermoplastic bonding layer; and wherein the bonding temperature comprises a melting temperature for the thermoplastic bonding layer.
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